

Literature Survey: APT Malware Detection Using Process Profiles

1. Main Idea of the Approach

The approach under review focuses on detecting APT malware based on how processes behave inside the operating system rather than by examining the malware files directly. Every program, whether benign or malicious, runs as a process. Malware inevitably creates or manipulates processes, leaving traces behind. By collecting process events, building them into process profiles represented as graphs, and then analyzing these graphs using advanced machine learning techniques, the approach enables early detection of APT malware.

The workflow can be summarized as: Process Events → Process Profile (Graph) → Graph2Vec Embedding → LSTM/BiLSTM Classification → Malware or Benign.

2. Methodology

2.1 Collecting Process Events

The first step involves using Sysmon, a Windows monitoring tool, to collect detailed process-related events from the operating system kernel. Key information includes process identifiers (ProcessID, Parent ProcessID), names, hash values, command-line arguments, executable image paths, creation and termination timestamps, and associated MITRE ATT&CK techniques. These details are standardized into structured records for each process.

2.2 Building Process Profiles

Processes are then represented as nodes in a directed graph, while parent-child relationships are represented as edges. Over time, this produces a process profile in the form of a graph that records how processes were created and related. Additionally, MITRE ATT&CK mappings provide context for each process, labeling them with behaviors such as file access, registry modifications, or network communication.

2.3 Graph2Vec Embedding

Since graphs are difficult to compare directly, each process profile is converted into a fixed-length numeric vector using Graph2Vec. The procedure works by decomposing each graph into rooted subgraphs, hashing them into unique identifiers, and then treating these subgraphs as words in a document. Graph2Vec applies a doc2vec-style training (PV-DBOW with skip-gram) to learn a vector embedding (64, 128, or 256 dimensions) for each graph. The result is a compact vector fingerprint that captures the structural and behavioral properties of the process profile.

2.4 Classification Using Deep Learning

The embeddings generated by Graph2Vec are input into Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) networks. LSTM networks are capable of remembering temporal patterns across sequences, which is important for analyzing process behavior. BiLSTM extends this by processing sequences in both forward and backward directions, allowing the network to capture richer context. A final Softmax regression layer outputs the classification decision: whether the process profile is malicious (APT malware) or benign.

3. Experimental Setup

The experimental dataset consisted of approximately 69,000 APT malware samples (from families such as Emotet, Dridex, NjRAT, etc.) and 288,000 benign files (including executables, DLLs, Office documents, PDFs, scripts, archives, and URLs). This resulted in a dataset of around 358,000 samples. The dataset was split into 80% training and 20% testing subsets.

The experiments compared the proposed Graph2Vec + LSTM/BiLSTM method against other approaches such as Random Forest, Multilayer Perceptron (MLP), Convolutional Neural Networks (CNNs), and Sequence-based models. Evaluation metrics included Accuracy, Precision, Recall, and F1-score. Experiments were implemented in Python 3.6 with TensorFlow 2.0 and Gensim 3.6.0 on Ubuntu 18.04, using an Intel Xeon CPU and an NVIDIA Tesla T4 GPU.

4. Results

The results showed that Graph2Vec combined with LSTM or BiLSTM consistently outperformed other approaches. Graph2Vec + LSTM achieved approximately 98% accuracy, with recall rates up to 96.5%, making it effective at detecting malware samples while maintaining computational efficiency. Graph2Vec + BiLSTM achieved slightly higher accuracy (around 98.4%) but required nearly twice the training time compared to LSTM. Other methods such as Random Forest, MLP, and CNNs performed reasonably well but had lower recall or higher false positives.

5. Key Insights

1. Malware may change its file structure, but it cannot avoid creating processes to execute. 2. Individual processes may appear harmless, but their sequence and relationships reveal hidden malicious behavior. 3. Process profiles modeled as graphs capture these correlations effectively. 4. Graph2Vec embeddings provide compact and meaningful representations of these graphs. 5. LSTM and BiLSTM classifiers are well-suited for identifying sequential and contextual patterns in the embeddings.

6. Conclusion

This literature survey highlights a promising approach for detecting APT malware through process profile analysis. By leveraging Sysmon for data collection, Graph2Vec for embedding

process graphs, and LSTM/BiLSTM for classification, the method achieves high accuracy and recall, outperforming traditional detection techniques. This approach opens new directions for detecting not only APT malware but also other types of anomalous behaviors on workstations.